Yakeen Plus (2025)

Practice Test - 08

DURATION :200 Minutes | DATE : 13/11/2024 | M.MARKS :720

ANSWER KEY

PHYSICS		CHEMISTRY		BOTANY		ZOOLOGY	
1.	(1)	51.	(1)	101.	(4)	151.	(2)
2.	(2)	52.	(1)	102.	(3)	152.	(2)
3.	(3)	53.	(3)	103.	(4)	153.	(3)
4.	(4)	54.	(1)	104.	(3)	154.	(3)
5.	(1)	55.	(2)	105.	(2)	155.	(3)
6.	(2)	56.	(3)	106.	(2)	156.	(1)
7.	(4)	57.	(1)	107.	(4)	157.	(4)
8.	(4)	58.	(3)	108.	(4)	158.	(4)
9.	(2)	59.	(4)	109.	(4)	159.	(3)
10.	(4)	60.	(3)	110.	(2)	160.	(1)
11.	(2)	61.	(1)	111.	(3)	161.	(2)
12.	(1)	62.	(2)	112.	(4)	162.	(3)
13.	(2)	63.	(2)	113.	(1)	163.	(4)
14.	(1)	64.	(3)	114.	(2)	164.	(4)
15.	(1)	65.	(1)	115.	(1)	165.	(3)
16.	(3)	66.	(1)	116.	(4)	166.	(2)
17.	(2)	67.	(1)	117.	(2)	167.	(1)
18.	(4)	68.	(3)	118.	(2)	168.	(2)
19.	(2)	69.	(3)	119.	(2)	169.	(3)
20.	(2)	70.	(2)	120.	(3)	170.	(1)
21.	(2)	71.	(3)	121.	(1)	171.	(2)
22.	(3)	72.	(1)	122.	(4)	172.	(3)
23.	(2)	73.	(1)	123.	(1)	173.	(3)
24.	(4)	74.	(3)	124.	(4)	174.	(2)
25.	(4)	75.	(1)	125.	(3)	175.	(3)
26.	(4)	76.	(3)	126.	(2)	176.	(4)
27.	(2)	77.	(2)	127.	(4)	177.	(4)
28.	(1)	78.	(1)	128.	(2)	178.	(3)
29.	(3)	79.	(2)	129.	(4)	179.	(2)
30.	(3)	80.	(2)	130.	(4)	180.	(4)
31.	(2)	81.	(3)	131.	(3)	181.	(1)
32.	(2)	82.	(4)	132.	(4)	182.	(3)
33.	(2)	83.	(2)	133.	(4)	183.	(2)
34.	(4)	84.	(1)	134.	(1)	184.	(2)
35.	(2)	85.	(3)	135.	(3)	185.	(4)
36.	(4)	86.	(4)	136.	(4)	186.	(2)
37.	(3)	87.	(1)	137.	(3)	187.	(4)
38.	(3)	88.	(2)	138.	(1)	188.	(4)
39.	(4)	89.	(2)	139.	(2)	189.	(1)
40.	(1)	90.	(2)	140.	(1)	190.	(3)
41.	(1)	91.	(4)	141.	(3)	191.	(3)
42.	(1)	92.	(2)	142.	(4)	192.	(1)
43.	(2)	93.	(4)	143.	(2)	193.	(2)
44.	(1)	94.	(2)	144.	(1)	194.	(3)
45.	(4)	95.	(2)	145.	(3)	195.	(3)
46.	(2)	96.	(1)	146.	(3)	196.	(3)
47.	(4)	97.	(4)	147.	(2)	197.	(1)
48.	(2)	98.	(3)	148.	(4)	198.	(1)
49.	(2)	99.	(3)	149.	(4)	199.	(3)
50.	(1)	100.	(3)	150.	(3)	200.	(3)

SECTION-(I) PHYSICS

1. **(1)**

$\delta^\circ = 360^\circ - 2\theta$

$180^\circ = 360^\circ - 2\theta$

$\Rightarrow 2\theta = 180^\circ \Rightarrow (\theta = 90^\circ)$

[New NCERT Class 12th Page No. 222]

2. **(2)**

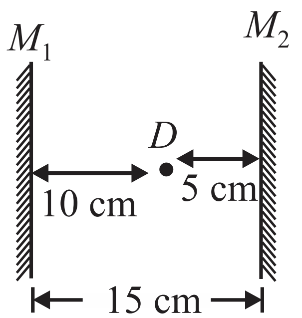
Condition for TIR

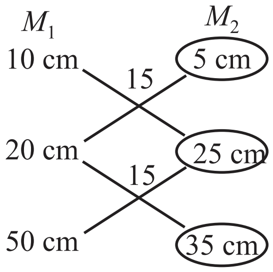
(i) $\theta > \theta_C$

(ii) Light must travel from denser medium to rarer medium.

[New NCERT Class 12th Page No. 230]

3. **(3)**





[New NCERT Class 12th Page No. 222]

4. **(4)**

$$f = \frac{d^2 - x^2}{4d}$$

$d = 80 + 40 = 120$

$x = 80 - 40 = 40$

$$f = \frac{120^2 - 40^2}{4 \times 120} = \frac{80}{3}\text{cm}$$

[New NCERT Class 12th Page No. 234]

5. **(1)**

$$\frac{1}{f} = \left({}_g\mu_a - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \left(\frac{2}{3} - 1\right)\left(\frac{2}{10}\right)$$

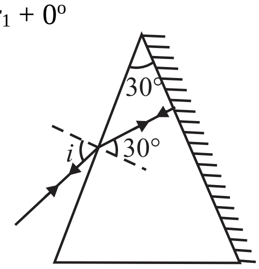
$\Rightarrow \quad f = -15$ cm, so behaves as concave lens.

[New NCERT Class 12th Page No. 234]

6. **(2)**

$A = r_1 + r_2$

$\therefore\ 30^o = r_1 + 0^o$



or $r_1 = 30^o$

Now, $\dfrac{\sin i}{\sin 30^o} = \sqrt{2}$

or $\sin i = \sqrt{2} \times \dfrac{1}{2}$

or $\sin i = \dfrac{1}{\sqrt{2}}$

or $i = 45^o$

[New NCERT Class 12th Page No. 240]

7. **(4)**

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\frac{A}{2}}$$

For equilateral prism, $\angle A = 60^o$

$$\mu = \frac{\sin\left(\frac{60^o + 60^o}{2}\right)}{\sin\frac{60^o}{2}} = \frac{\sin 60^o}{\sin 30^o} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3} = 1.73$$

[New NCERT Class 12th Page No. 240]

8. **(4)**

$\delta = (360 - 2\theta) = (360 - 2 \times 60) = 240^\circ$

120° deviation is also

[New NCERT Class 12th Page No. 241]

9. **(2)**

In each case two plane-convex lens are placed close to each other, and $\dfrac{1}{F} = \dfrac{1}{f_1} + \dfrac{1}{f_2}$.

[New NCERT Class 12th Page No. 238]

10. **(4)**

Number of images = (Number of materials)

[New NCERT Class 12th Page No. 222]

11. **(2)**

For total internal reflection the angle of incidence should be greater than the critical angle. As critical angle is approximately 45°. Therefore, total internal reflection is not possible. So, assertion is not true but reason is true.

[New NCERT Class 12th Page No. 230]

12. **(1)**

μ depends on colour of light.

$\mu \propto \frac{1}{\lambda}$

and does not depend on the prism angle and deviation of light.

[New NCERT Class 12th Page No. 240]

13. **(2)**

When object is placed, between focus and pole, image formed is erect, virtual and enlarged.

[New NCERT Class 12th Page No. 224]

14. **(1)**

By formula $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$= (1.5 - 1)\left(\frac{1}{40} + \frac{1}{40}\right) = 0.5 \times \frac{1}{20} = \frac{1}{40}$

$\therefore \quad f = 40$ cm

[New NCERT Class 12th Page No.235]

15. **(1)**

By using formula

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$\Rightarrow \quad \frac{1.5}{v} - \frac{1}{(-15)} = \frac{(1.5 - 1)}{+30}$

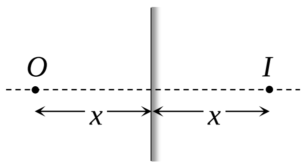
$\Rightarrow \quad v = -30$ cm.

Negative sign shows that, image is obtained on the same side of object *i.e.*, towards left.

[New NCERT Class 12th Page No. 232]

16. **(3)**

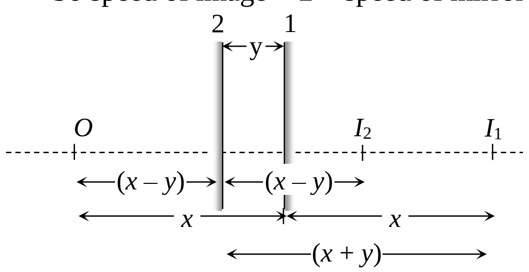
Suppose at any instant, plane mirror lies at a distance x from object. Image will be formed behind the mirror at the same distance x.



When the mirror shifts towards the object by distance *'y'* the image shifts

$= x + y - (x - y) = 2y$

So speed of image = 2 × speed of mirror



[New NCERT Class 12th Page No. 222]

17. **(2)**

Nature of lens changes, if $\mu_{medium} > \mu_{lens}$

[New NCERT Class 12th Page No. 234]

18. **(4)**

$f = \frac{R}{2(\mu - 1)}$

$f' = \frac{R}{(\mu - 1)}$

$\Rightarrow \quad f' = 2f$

[New NCERT Class 12th Page No. 235]

19. **(2)**

Resonance frequency

$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8 \times 10^{-3} \times 20 \times 10^{-6}}} = 2500\ rad/\text{sec}$

Resonance current = $\frac{V}{R} = \frac{220}{44} = 5\text{A}$

[New NCERT Class 12th Page No. 187 to 189]

20. **(2)**

$R = \frac{P}{i_{\text{rms}}^2} = \frac{240}{16} = 15\,\Omega$

$Z = \frac{V}{i} = \frac{100}{4} = 25\,\Omega$

Now $X_L = \sqrt{Z^2 - R^2} = \sqrt{(25)^2 - (15)^2} = 20\,\Omega$

$\therefore 2\pi v L = 20 \Rightarrow L = \frac{20}{2\pi \times 50} = \frac{1}{5\pi}\text{Hz}$

[New NCERT Class 12th Page No. 190, 191]

21. **(2)**

At $t = 0$, phase of the voltage is zero, while phase of the current is $-\frac{\pi}{2}$ i.e., voltage leads by $\frac{\pi}{2}$

[New NCERT Class 12th Page No. 187 to 189]

22. **(3)**

$V_{\text{rms}} = \frac{200}{\sqrt{2}},\ i_{\text{rms}} = \frac{1}{\sqrt{2}}$

$\therefore P = V_{\text{rms}}\, i_{\text{rms}} \cos\phi = \frac{200}{\sqrt{2}} \frac{1}{\sqrt{2}} \cos\frac{\pi}{3} = 50$ watt

[New NCERT Class 12th Page No. 190, 191]

23. **(2)**

$Q = \frac{\text{Resonant frequency}}{\text{Band width}}$

$\Delta\omega \Rightarrow \frac{600}{4} = 150$ Hz

[New NCERT Class 12th Page No. 187 to 189]

24. **(4)**

Current $I=\frac{E}{Z}$ where $E=\sqrt{V_R^2+\left(V_L-V_C\right)^2}$

$Z=\sqrt{R^2+\left(X_L-X_C\right)^2}$

At resonance, $X_L=X_C \quad \therefore\ Z=R$

Again at resonance, $V_L=V_C \quad \therefore\ E=V_R$

$\therefore \quad I=\frac{V_R}{R}=\frac{100}{1\times10^3}=0.1 \text{ A}$

$\therefore \quad V_L=IL\omega=\frac{I}{C\omega}=\frac{0.1}{(2\times10^{-6})\times(200)}$

$\therefore \quad V_L=250 \text{ volt}$

[New NCERT Class 12th Page No. 187 to 189]

25. **(4)**

$\sqrt{\frac{V_0^2+0^2}{2}}=\frac{V_0}{\sqrt{2}}$

[New NCERT Class 12th Page No. 178, 179]

26. **(4)**

In an *LCR* series a.c. circuit, the voltages across components *L* and *C* are in opposite phase. The voltage across *LC* combination will be zero.

[New NCERT Class 12th Page No. 187 to 189]

27. **(2)**

$Q\text{-fact}=\frac{1}{R}\sqrt{\frac{L}{C}}=\frac{1}{10\times10^3}\sqrt{\frac{0.8}{0.2\times10^{-6}}}$

$=\frac{2\times10^3}{10\times10^3}=0.2$

[New NCERT Class 12th Page No. 187 to 189]

28. **(1)**

L.C.R Circuit:

Because the phase difference between current and voltage in the case of a resistor is zero, energy dissipates only through resistance in an L.C.R circuit.

[New NCERT Class 12th Page No. 187 to 189]

29. **(3)**

$20=100\pi L$

$\frac{1}{5\pi}=L$

$X_L=\omega L=2\pi(100)\left(\frac{1}{5\pi}\right)=40\Omega$

and $R=30\Omega$

$\Rightarrow Z=\sqrt{R^2+X_L^2}=\sqrt{(30)^2+(40)^2}=50\Omega$

So $I=\frac{200}{50}=4 \text{ A}$

[New NCERT Class 12th Page No. 187 to 189]

30. **(3)**

Since the circuit is in resonance.

So, $i=\frac{220}{100}=2.2\text{A}$

[New NCERT Class 12th Page No. 181 to 183]

31. **(2)**

We know that

$I_{rms}=I_0/\sqrt{2}$ and $I_\text{m}=2I_0/\pi$

$\therefore \frac{I_\text{m}}{I_{rms}}=\frac{2\sqrt{2}}{\pi}$

[New NCERT Class 12th Page No. 178, 179]

32. **(2)**

$X_L=R, X_C=R/2$

$\therefore\ \tan\phi=\frac{X_L-X_C}{R}=\frac{R-\frac{R}{2}}{R}=\frac{1}{2}$

$\Rightarrow \phi=\tan^{-1}(1/2)$

Also $Z=\sqrt{R^2+(X_L-X_C)^2}=\sqrt{R^2+\frac{R^2}{4}}=\frac{\sqrt{5}}{2}R$

[New NCERT Class 12th Page No. 187 to 189]

33. **(2)**

Here current, $i=i_1\sin\omega t+i_2\cos\omega t$

$\Rightarrow\ i=\sqrt{i_1^2+i_2^2}\left[\frac{i_2\cos\omega t}{\sqrt{i_1^2+i_2^2}}+\frac{i_1\sin\omega t}{\sqrt{i_1^2+i_2^2}}\right]$

$\Rightarrow\ i=\sqrt{i_1^2+i_2^2}\ [\sin\alpha\cos\omega t+\cos\alpha\sin\omega t]$

$\Rightarrow\ i=\sqrt{i_1^2+i_2^2}\ \sin(\omega t+\alpha)=i_0\sin(\omega t+\alpha)$

$\Rightarrow\ i_{rms}=\frac{i_0}{\sqrt{2}}=\sqrt{\frac{i_1^2+i_2^2}{2}}$

[New NCERT Class 12th Page No. 187 to 189]

34. **(4)**

R and *L* cause phase difference to lie between 0 and π/2 but never 0 and π/2 at extremities.

[New NCERT Class 12th Page No. 187 to 189]

35. **(2)**

Here, Efficiency of the transformer, η = 90%

Input power, $P_{in}=3\,\text{kW}=3\times10^3\text{ W}=3000\text{ W}$

Voltage across the primary coil, $V_p=200\text{ V}$

Current in the secondary coil, $I_s=6\text{ A}$

As, $P_{in}=I_pV_p$

∴ Current in the primary coil.

$I_P=\frac{P_{in}}{V_P}=\frac{3000W}{200V}=15\text{ A}$

Efficiency of the transformer,

$\eta=\frac{P_{out}}{P_{in}}=\frac{V_sI_s}{V_pI_p}$

$\therefore\ \frac{90}{100}=\frac{6V_s}{3000}$ or $V_s=\frac{90\times3000}{100\times6}=450\text{ V}$

[New NCERT Class 12th Page No. 190, 191]

36. **(4)**

Here, Input voltage, V_P = 220 V

Output voltage, V_s = 440 V

Input current, I_p = ?

Output current, I_s = 2 A

Efficiency of the transformer, η = 80%

Efficiency of the transformer,

$$\eta = \frac{\text{Output power}}{\text{Input power}}$$

$$\eta = \frac{V_s I_s}{V_p I_p} \Rightarrow I_p = \frac{V_s I_s}{\eta V_p} = \frac{(440\text{V})(2\text{A})}{\left(\frac{80}{100}\right)(220\text{V})}$$

$$= \frac{(440\text{V})(2\text{A})(100)}{(80)(220\text{V})} = 5\text{A}$$

[New NCERT Class 12th Page No. 190, 191]

37. **(3)**

The electromagnetic waves of all wavelengths travel with the same speed in space which is equal to velocity of light.

[New NCERT 12th Page No. 310]

38. **(3)**

Conceptual

[New NCERT 12th Page No. 312]

39. **(4)**

Light wave is an electromagnetic wave in which $\vec{E}$ and $\vec{B}$ are at right angles to each other as well as at right angles to the direction of wave propagation.

[New NCERT 12th Page No. 316]

40. **(1)**

Velocity of light

$$C = \frac{E}{B} \Rightarrow B = \frac{E}{C} = \frac{9.3}{3\times10^8} = 3.1 \times 10^{-8}\text{ T}$$

[New NCERT 12th Page No. 311]

41. **(1)**

$\frac{f_0}{f_e} = 9$ $\quad\therefore\quad f_0 = 9f_e$

Also $f_0 + f_e = 20$ (∵ final image is at infinity)

$9f_e + f_e = 20$, $f_e = 2$ cm

∴ $f_0 = 18$ cm

[New NCERT 12th Page No. 319]

42. **(1)**

Here, $\lambda = \frac{c}{v} = \frac{3\times10^8}{8.2\times10^6} = 36.6$ m

[New NCERT 12th Page No. 312]

43. **(2)**

Angle between plane of vibration and plane of polarisation is 90°.

[New NCERT 12th Page No. 360]

44. **(1)**

Angle between plane of polarisation and direction of propagation is 0°.

[New NCERT 12th Page No. 362]

45. **(4)**

Conceptual

[New NCERT 12th Page No. 367]

46. **(2)**

Conceptual

[New NCERT 12th Page No. 371]

47. **(4)**

Intensity I ∝ 1 + cos ϕ where ϕ = phase difference

$$\therefore \frac{I_P}{I_Q} = \frac{1+\cos 0^\circ}{1+\cos\frac{\pi}{2}} = \frac{1+1}{1+0} = \frac{2}{1}$$

[New NCERT 12th Page No. 355]

48. **(2)**

The intensity I is given as

$I = I_0 \cos^2\frac{\phi}{2}$ where I_0 is the peak intensity

Here $I = \frac{I_0}{2}$, $\therefore \frac{I_0}{2} = I_0\cos^2\frac{\phi}{2} \therefore \phi = \frac{\pi}{2}(2n+1)$

For a phase difference of 2π the path difference is λ

∴ For a phase difference of $(2n+1)\frac{\pi}{2}$ the path difference is $(2n+1)\frac{\lambda}{4}$. option (2) is correct

[New NCERT 12th Page No. 356]

49. **(2)**

The position of n[th] dark fringe $= \frac{n\lambda D}{2d}$. So position of first dark fringe in $x_1 = \lambda D/2d$.

D = 20 cm, d = 0.1mm, λ = 5460 Å, θ = 0.16°

[New NCERT 12th Page No. 360]

50. **(1)**

$a\sin\theta = n\lambda$ $\qquad \frac{ax}{f} = 3\lambda$

(since θ is very small so sinθ ≈ tanθ ≈ θ = x/f)

or $\lambda = \frac{ax}{3f} = \frac{0.3\times10^{-3}\times5\times10^{-3}}{3\times1}$

$= 5\times10^{-7}$ m = 5000 Å.

[New NCERT 12th Page No. 362]

SECTION-(II) CHEMISTRY

51. **(1)**
All have D configuration.
[New NCERT Class 12th Page No. 286]

52. **(1)**
Both Glucose and fructose have the same molecular formula but different functional groups. Glucose is an aldohexose (aldehyde group) and fructose is a ketohexose (ketonic group). Hence they are functional isomers.
[New NCERT Class 12th Page No. 284]

53. **(3)**
Methionine contains a sulfur atom in its side chain ($-CH_2-CH_2-S-CH_3$)
[New NCERT Class 12th Page No. 291]

54. **(1)**
They are anomers.
[New NCERT Class 12th Page No. 285]

55. **(2)**
To convert glucose to saccharic acid, the reagent used is nitric acid. Dilute nitric acid oxidises glucose to saccharic acid. Primary –OH group and –CHO group of glucose are oxidised to –COOH group.

$$\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} \xrightarrow{HNO_3} \begin{array}{c} COOH \\ | \\ (CHOH)_4 \\ | \\ COOH \end{array}$$

Glucose Saccharic acid
[New NCERT Class 12th Page No. 284]

56. **(3)**
Denaturation does not change the primary structure of protein.
[New NCERT Class 12th Page No. 294]

57. **(1)**
The phosphodiester linkage in nucleotides is present between the 5' carbon atom of one pentose sugar and the 3' carbon atom of the next pentose sugar.
[New NCERT Class 12th Page No. 285]

58. **(3)**
- Anemia is proteins affected disease
- Rickets is vitamin D deficiency disease.
- Pyorrhea is known as periodontitis, is a multifactorial disease affecting oral gum. It is an inflammation caused by bacteria.
- Beri-Beri is caused by deficiency of vitamin B_1.

[New NCERT Class 12th Page No. 291]

59. **(4)**
Terminal alkynes give white precipitate with ammoniacal silver nitrate solution.
[Newly Added Topic]

60. **(3)**
Aliphatic and aromatic amine can be differentiated by diazotization followed by β-naphthol.
[Newly Added Topic]

61. **(1)**
Tollen's reagent gives a positive test with all aldehyde.
[Newly Added Topic]

62. **(2)**
Alcohols, Alkyl benzenes and alkene give reaction with acidic $KMnO_4$.
[Newly Added Topic]

63. **(2)**
Acid produces CO_2 on reaction with $NaHCO_3$.
[Newly Added Topic]

64. **(3)**
$\left[Fe(CN)_6\right]^{3-}$: Cyanide (CN^-) is a strong field ligand. This complex will have a significantly higher Δ_o value compared to the other complexes as water is a weak field ligand.
[New NCERT Class 12th Page no. 133]

65. **(1)**
Primary valencies are ionisable and secondary valencies are non-ionisable.
[New NCERT Class 12th Page No. 120]

66. **(1)**
- In coordination compounds metals show two types of linkages (valencies)-primary and secondary.
- The primary valences are normally ionisable and are satisfied by negative ions.
- The secondary valences are non ionisable. These are satisfied by neutral molecules or negative ions called ligands.

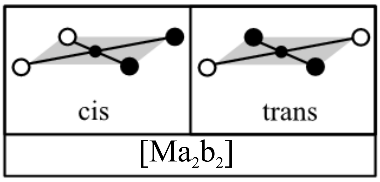
[New NCERT Class 12th Page No. 119]

67. **(1)**
$\left[MnO_4\right]^-$ has Mn in +7 oxidation state.
Mn^{7+} : [Ar]. It has no d-elctrons.
[New NCERT Class 12th Page No. 120]

68. **(3)**

In the given options, M represents metal while 'a' and 'b' are monodentate ligands It is seen that cis and trans isomerism occurs only in octahedral and square planar complexes and not in tetrahedral.

Cis configuration arises when two same ligands are adjacent to each other while trans arises when they are opposite to each other. Out of the following only can have this type of geometry.



[New NCERT Class 12th Page No. 125]

69. **(3)**

Acid reacts with $NaHCO_3$ while ester does not react with $NaHCO_3$.

[Newly Added Topic]

70. **(2)**

Aromatic aldehyde do not give a positive fehling's test.

[Newly Added Topic]

71. **(3)**

Compund having unsaturation will decolourise $Br_2–H_2O$ and Enols give positive test with neutral $FeCl_3$.

[Newly Added Topic]

72. **(1)**

Aldehydes give the positive test with Tollen's reagent.

[Newly Added Topic]

73. **(1)**

Paramagnetism $\propto$ Number of unpaired electrons.

$\left[Mn\left(H_2O\right)_6\right]^{2+}$ has Mn in +2 oxidation state and water is a weak field ligand. So, total 5 unpaired electrons are present. So maximum Para magnetism shown by $\left[Mn\left(H_2O\right)_6\right]^{2+}$ complex ion.

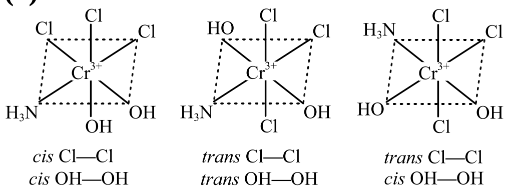
[New NCERT Class 12th Page No. 120]

74. **(3)**

VBT does not explain the colour exhibited by co-ordination compounds. VBT does not explains and gives quantitative interpretation of magnetic data. VBT does not distinguish between weak field ligand and strong field ligand.

[New NCERT Class 12th Page No. 131]

75. **(1)**



[New NCERT Class 12th Page No. 126]

76. **(3)**

(i) The dissociation of complex $\left[Co\left(NH_3\right)_6\right]Cl_2$ in aquoeus solution is as follows;

$$\left[Co\left(NH_3\right)_6\right]Cl_2 \rightleftharpoons \left[Co\left(NH_3\right)_6\right]^{2+} + 2Cl^-$$

(ii) The number of ions formed is 3

[New NCERT Class 12th Page No. 119]

77. **(2)**

Coordination number of Fe in K_3 $[Fe(CN)_6]$ is 6 because it is bonded with six $\bar{C}N$ ligands. Let x be the oxidation state of Fe

$\therefore\ x + 6(-1) = -3x = -3$

$\therefore\ x = +3$

[New NCERT Class 12th Page No. 122]

78. **(1)**

$$\mu = \sqrt{n\left(n+2\right)} = 2.84 \Rightarrow n = 2$$

Electronic configuration of $Ni^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$ having two unpaired electrons.

Electronic configuration of $Ti^{3+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$ having one unpaired electron.

Electronic configuration of $Cr^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^4$ having four unpaired electrons.

Electronic configuration of $Co^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$ having three unpaired electrons.

[New NCERT 12th Page No. 102]

79. **(2)**

Within a transition series, the ionic radius generally decreases with increasing atomic number. This is due to the increasing nuclear charge and the imperfect shielding effect of d electrons.

[New NCERT Class 12th Page no. 99]

80. **(2)**

Electronic configuration of Mn is $\left[Ar\right]3d^5 4s^2$

With oxidation numbers of +2, +3, +4, +6 and +7, manganese exhibits the highest oxidation state.

[New NCERT Class 12th Page no. 96]

81. **(3)**

$\left[Xe\right]4f^{0-14}5d^{1-10}6s^{1-2}$

[New NCERT 12th Page No. 110]

82. **(4)**

They can show variable oxidation states.

[New NCERT Class 12th Page No. 104]

83. **(2)**
The most characteristic oxidation state of lanthanides is +3.
[New NCERT Class 12th Page No. 109]

84. **(1)**
Cu^{2+} ion is more stable than Cu^{+} ion due to high charge density on Cu^{2+} ion, it has higher hydration energy when it bonds to water molecules.
$\therefore (A)$ is correct but (R) is not correct.
[New NCERT Class 12th Page No. 100]

85. **(3)**
Oxidized product can not be SO_4^{2-}.
[New NCERT Class 12th Page No. 106]

86. **(4)**
Percentage of lanthanoid in Mischmetal is 95%.
[New NCERT Class 12th Page No. 113]

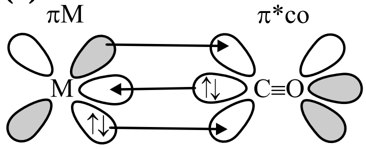
87. **(1)**
Increasing order of melting point:
Zn < Cu < Ni < Fe
[New NCERT Class 12th Page No. 95]

88. **(2)**
It has a tendency to attain noble gas configuration.
[New NCERT Class 12th Page No. 112]

89. **(2)**
Pink colouration in Schiff's test indicates the presence of aldehyde.
[Newly Added Topic]

90. **(2)**
Yellow colour of precipitate formed in iodoform test.
[Newly Added Topic]

91. **(4)**
Lassaigne's test is not used for the detection of oxygen.
[Newly Added Topic]

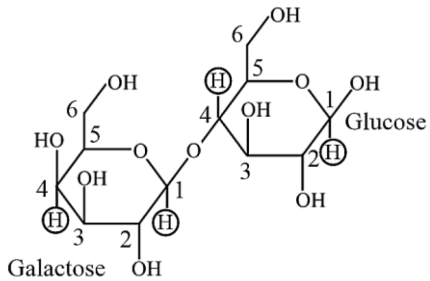
92. **(2)**


[New NCERT Class 12th Page No. 135]

93. **(4)**
For IUPAC naming of complexes, ligands are written in alphabetical order followed by metal ion in its respective oxidation state. Primary valencies are shown outside the coordination sphere.
$\therefore$ IUPAC Name of the given compound is:
Tetraammineaquachloridocobalt(III) chloride
[New NCERT Class 12th Page No. 124]

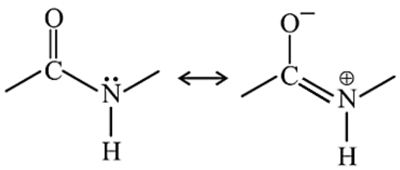
94. **(2)**
Ligands which are capable of accepting an appreciable amount of π electron density from the metal atom into π or π^* orbitals of their own are called π-acceptor or π-acid ligands such as CO.
[New NCERT Class 12th Page No. 135]

95. **(2)**
This compound contains two oxygen donor atoms and two nitrogen donor atoms
So, it will be a tetradentate ligand
[New NCERT Class 12th Page No. 121]

96. **(1)**
Lactose is a disaccharide sugar


In lactose C_1–β of galactose is attached to C_4 of glucose.
[New NCERT Class 12th Page No. 288]

97. **(4)**
The phenomenon where alpha and beta glucose, with different specific rotations, convert into each other in an aqueous solution until reaching a constant specific rotation is called mutarotation.
[New NCERT Class 12th Page No. 285]

98. **(3)**
Peptide linkage is an amide linkage, has partial double bond character due to resonance.

Protein chains are associated by H-bond, it is hydrophilic (water loving).
[New NCERT Class 12th Page No. 292]

99. **(3)**
Cellulose is composed of β-D–glucose units which are joined by glycosidic linkage between C-1 of one glucose unit and C-4 of next glucose unit.
[New NCERT Class 12th Page No. 289]

100. **(3)**
In α-helix structure of protein, a polypeptide chain is stabilized by the formation of intramolecular H-bonds between –NH– group of amino acids in one turn with the C = O groups of amino acids belonging to adjacent turn.
[New NCERT Class 12th Page No. 299]

SECTION-(III) BOTANY

101. (4)

LAB is used to prepare curd.

LAB improves milk nutritional quality by increasing vitamin B_{12}.

LAB prevents disease causing microbes in gut.

[New NCERT Class 12th Page No. 151]

102. (3)

Parent	:	Rr	×	rr
		(Pink)		(White)
Gametes	:	R, r		r
Progeny	:	Rr : rr		
Ratio	:	1 Pink : 1 White		

[New NCERT Class 12th Page No. 60]

103. (4)

If three genes A, B and C control skin colour in human with the dominant forms A, B and C responsible for dark skin colour and the recessive forms a, b and c for light skin colour. The genotype with all the dominant alleles (AABBCC) will have the darkest skin colour and that with all the recessive alleles (aabbcc) will have the lightest skin colour.

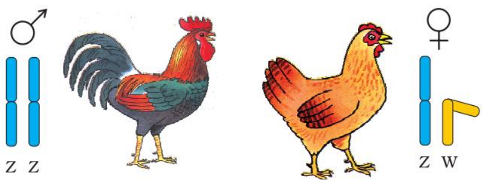
[New NCERT Class 12th Page No. 69]

104. (3)

Since there are three different alleles, there are six different combinations of these three alleles that are possible, and therefore, a total of six different genotypes of the human ABO blood types.

[New NCERT Class 12th Page No. 61]

105. (2)



(P) **(Q)**

P is male and has a pair of similar chromosomes ZZ and Q is female and has two dissimilar ZW chromosomes.

[New NCERT Class 12th Page No. 70]

106. (2)

Mendelian disorders are mainly determined by alteration or mutation in the single gene.

[New NCERT Class 12th Page No. 74]

107. (4)

- Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are the two types of nucleic acids found in living systems.
- DNA acts as the genetic material in most of the organisms.

[New NCERT Class 12th Page No. 70]

108. (4)

The nutritive medium for growing bacteria and many fungi in the laboratory is called culture media.

[New NCERT Class 12th Page No. 149]

109. (4)

Trichoderma polysporum	Cyclosporin A
Monascus purpureus	Statins
Streptococcus	Streptokinase
Clostridium butylicum	Butyric acid

[New NCERT Class 12th Page No. 153]

110. (2)

Azospirillum is a free living bacterium that plays an important role in nitrogen fixation in the soil.

[New NCERT Class 12th Page No. 158]

111. (3)

The primary treatment of sewage involves filtration and sedimentation. The secondary treatment involves the breakdown of the organic matter with the help of microbes. Therefore, microbial proliferation is required during the secondary treatment of the sewage.

[New NCERT Class 12th Page No. 154]

112. (4)

Glomus help in absorption of phosphorus from soil by plants.

[New NCERT Class 12th Page No. 158]

113. (1)

The Ministry of Environment and Forests has initiated Ganga Action Plan and Yamuna Action Plan to save the major rivers of our country from pollution.

[New NCERT Class 12th Page No. 155]

114. (2)

- RNA is known to be catalytic and more reactive in comparison to DNA.
- RNA functions as adapter, structural, and in some cases as a catalytic molecule.

[New NCERT Class 12th Page No. 79]

115. (1)

Aspergillus niger	Citric acid
Acetobacter aceti	Acetic acid
Saccharomyces cerevisiae	Ethanol
Lactobacillus	Lactic acid

[New NCERT Class 12th Page No. 153]

116. (4)

Satellite DNA is important because it shows a high degree of polymorphism in the population and also the same degree of polymorphism in an individual, which is heritable from parents to children.

[New NCERT Class 12th Page No. 106]

117. (2)

Methane, CO_2 and hydrogen are the gases present in biogas.

[New NCERT Class 12th Page No. 155]

118. (2)

61 codons out of 64 codes for 20 different amino acids. It is referred to as degeneracy of genetic code.

[New NCERT Class 12th Page No. 96]

119. (2)

Thymine is present in DNA while in RNA there is uracil.

[New NCERT Class 12th Page No. 81]

120. (3)

Fleming, Chain and Florey were awarded the Nobel Prize in 1945, for the discovery.

[New NCERT Class 12th Page No. 152]

121. (1)

The supernatant left after sedimentation of sewage water during primary treatment is known as effluent.

[New NCERT Class 12th Page No. 154]

122. (4)

Bacteriophage $\phi \times 174$	5386 nucleotides
Bacteriophage lambda	48502 bp
Escherichia coli	4.6×10^6 bp
Haploid content of human DNA	3.3×10^9 bp

[New NCERT Class 12th Page No. 80]

123. (1)

Baculoviruses are pathogens that attack insects and other arthropods. The majority of baculoviruses used as biological control agents are in the genus *Nucleopolyhedrovirus*. These viruses are excellent candidates for species-specific, narrow spectrum insecticidal applications. They have been shown to have no negative impacts on plants, mammals, birds, fish or even on non-target insects. This is especially desirable when beneficial insects are being conserved to aid in an overall integrated pest management (IPM) programme, or when an ecologically sensitive area is being treated.

[New NCERT Class 12th Page No. 157]

124. (4)

- The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called a nucleosome in nucleus in eukaryotes.

[New NCERT Class 12th Page No. 83]

125. (3)

If the sequence of bases in one strand of DNA is ATGCAAGGA, the sequence of bases on complementary strand will be TACGTTCCT.As A pairs with T and G pairs with C.

5'ATGCAAGGA3'

3'TACGTTCCT5'

[New NCERT Class 12th Page No. 81]

126. (2)

High biological oxygen demand in a waste water body means water is polluted.

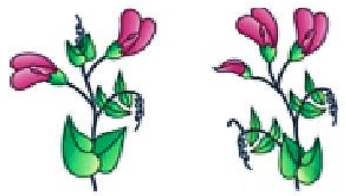
[New NCERT Class 12th Page No. 154]

127. (4)

- Mendel proposed two general rules to consolidate his understanding of inheritance in monohybrid crosses.
- Mendel self-pollinated the F_1 plants.
- The symbol ♂ is used to denote the male (pollens).
- The law of segregation is based on the fact that the alleles do not show blending.

[New NCERT Class 12th Page No. 57,59]

128. (2)

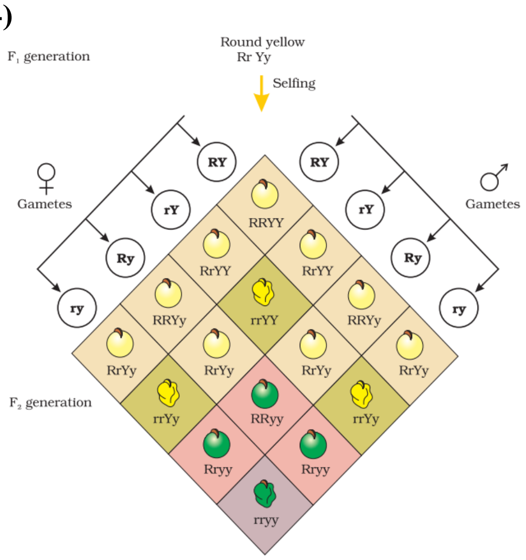


A **B**

A- Axial **B-** Terminal

[New NCERT Class 12th Page No. 54]

129. (4)



Phenotypic ratio : round yellow : round green : wrinkled yellow : wrinkled green

[New NCERT Class 12th Page No. 63]

130. (4)

Starch is synthesised effectively by BB homozygotes and therefore, large starch grains are produced. In contrast, bb homozygotes have lesser efficiency in starch synthesis and produce smaller starch grains. After maturation of the seeds, BB seeds are round and the bb seeds are wrinkled. Heterozygotes produce round seeds, and so B seems to be the dominant allele. But, the starch grains produced are of intermediate size in Bb seeds.

[New NCERT Class 12th Page No. 62]

131. (3)

The genotype of the father's blood group can not be $I^A I^B$.

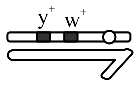
[New NCERT Class 12th Page No. 61]

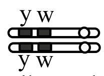
132. (4)

Antibiotics are chemical substances, which are produced by some microbes and can kill or retard the growth of other (disease-causing) microbes.

[NEW NCERT 12th Class Page No. 152]

133. (4)

- 

represents dominant wild type genotype in *Drosophila*.
- A cross between plants differing in one trait is called monohybrid cross and two traits is called dihybrid cross
- 

represents female with recessive
Yellow, white
linked genes in *Drosophila*.
- Chromosomes as well as genes occur in pairs.

[New NCERT Class 12th Page No. 68]

134. (1)

Parent: AaBb

Gametes: AB, Ab, aB, ab

The percentage of ab gamete produced by AaBb parent will be 25%.

[New NCERT Class 12th Page No. 62]

135. (3)

Biocontrol refers to the use of biological methods for controlling plant diseases and pests.

In agriculture, there is a method of controlling pests that relies on natural predation rather than introduced chemicals.

[NEW NCERT 12th Class Page No. 156-157]

136. (4)

The effluent from the secondary treatment plant is generally released into natural water bodies like rivers and streams.

The number of sewage treatment plants has not increased enough to treat sewage so sewage is often discharged directly into rivers leading to their pollution and increase in water-borne diseases.

[New NCERT Class 12th Page No. 155]

137. (3)

Colour blindness is X-linked recessive disorder.

[New NCERT Class 12th Page No.73]

138. (1)

RNA polymerase-III transcribes tRNA, 5SrRNA and snRNA.

[New NCERT Class 12th Page No.95]

139. (2)

4 genotypes- I^AI^A, I^AI^B, I^Bi and I^Ai.

3 phenotypes-A, B and AB.

[New NCERT Class 12th Page No.61]

140. (1)

Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.

[NEW NCERT 12th Class Page No. 153]

141. (3)

Red flower in snapdragon is a dominant trait.

[New NCERT Class 12th Page No. 60]

142. (4)

Tightly linked genes show lower recombination frequencies.

[New NCERT Class 12th Page No. 152]

143. (2)

Thymine residue- 0

Adenine residue- 5

[New NCERT Class 12th Page No.81,92 & 93]

144. (1)

These are available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants such as brassicas and fruit trees, where these are eaten by the insect larvae. In the gut of the larvae, the toxin is released and the larvae get killed. The bacterial disease will kill the caterpillars, but leave other insects unharmed.

[New NCERT Class 12th Page No. 157]

145. (3)

- Dominant- Violet
- Recessive- White
- Genotype-Vv for Violet
- Phenotype-White for vv

[New NCERT Class 12th Page No. 55, 56]

146. (3)

- Toddy, a traditional drink is made by fermenting sap from palms.
- Microbes are used to ferment bamboo shoots to make foods.

[New NCERT Class 12th Page No. 157]

147. (2)

Hershey-Chase experiment successfully proved DNA to be genetic material. The correct sequence of steps followed in this experiment is:

Infection → Blending → Centrifugation

[New NCERT Class 11th Page No. 86]

148. (4)

All genes that are expressed as RNA.

[New NCERT Class 11th Page No. 103]

149. (4)

The enzyme DNA ligase is used to join the discontinuously synthesised fragments during replication.

[New NCERT Class 11th Page No. 91]

150. (3)

The large holes in 'Swiss cheese' are due to production of a large amount of CO_2 by a bacterium named *Propionibacterium sharmanii.* The 'Roquefort cheese' are ripened by growing a specific fungi on them, which gives them a particular flavour.

[New NCERT Class 12th Page No. 151]

SECTION-(IV) ZOOLOGY

151. (2)

Amniocentesis is a technique that involves culturing cells and studying metaphase chromosomes from amniotic fluid to identify chromosomal abnormalities.

[New NCERT Class 12th Page No. 42]

152. (2)

The pill	Prevents ovulation
Vasectomy	Semen contains no sperms
Condom	Prevents sperms reaching cervix
Copper T	Reduce fertility capacity of sperms

[New NCERT Class 12th Page No. 44]

153. (3)

IVF is popularly known as test tube baby programme, ova from the wife/donor (female) and sperms from the husband/donor (male) are collected and are induced to form zygote under simulated conditions in the laboratory.

[New NCERT Class 12th Page No. 48]

154. (3)

First line of defense

Physical and chemical barriers that prevent pathogens from entering the body. These include the skin, mucous membranes, saliva, hair, and bodily excretions.

Third line of defense

Specific immune cells that target specific antigens. These include B-cells and T-cells, which are white blood cells. B-cells produce antibodies, and T-cells help identify and destroy pathogenic cells.

[New NCERT Class 12th Page No. 134]

155. (3)

- The government was forced to take up serious measures to check population growth rate.
- An ideal contraceptive should be user-friendly.

[New NCERT Class 12th Page No. 42, 43]

156. (1)

Statutory raising of marriageable age of the female to 18 years and that of males to 21 years.

[New NCERT Class 12th Page No. 43]

157. (4)

- Emergency contra captive pills may be taken up to 72 hours after coitus to prevent conception.
- Generally chances of conception are nil until mother breast-feeds the infant up to six months.

[New NCERT Class 12th Page No. 44]

158. (4)

The B-lymphocytes produce an army of proteins in response to pathogens into our blood to fight with them. These proteins are called antibodies.

[New NCERT Class 12th Page No. 130]

159. (3)

Oral administration of small doses of either progestogens or progestogen–estrogen combinations is another contraceptive method used by the females.

[New NCERT Class 12th Page No. 45]

160. (1)

- Progestogens alone or in combination with estrogen can also be used by females as injections or implants under the skin.
- Implants are used by females.

[New NCERT Class 12th Page No. 44, 45]

161. (2)

Well-known plants with hallucinogenic properties are *Atropa belladona* and *Datura*.

[New NCERT Class 12th Page No. 143]

162. (3)

Embryos with more than 8 blastomeres transfer into the uterus (IUT – intra uterine transfer).

[New NCERT Class 12th Page No. 48]

163. (4)

IUDs is one of most widely accepted methods of contraception in India.

[New NCERT Class 12th Page No. 45]

164. (4)

Artificial insemination means artificial introduction of sperms of a healthy donor into the vagina.

[New NCERT Class 12th Page No. 42]

165. (3)

- IUDs are ideal contraceptives for the females who want to delay pregnancy.
- Pills have to be taken daily for a period of 21 days starting preferably within the first five days of the menstrual cycle.

[New NCERT Class 12th Page No. 45]

166. (2)

The use of drugs like anti-histamine, adrenaline and steroids quickly reduce the symptoms of allergy.

[New NCERT Class 12th Page No. 137]

167. (1)

In tubectomy, a small part of the fallopian tube is tied or removed.

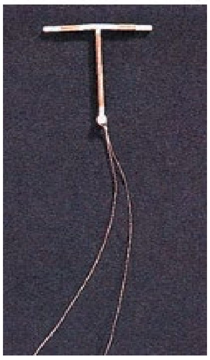
[New NCERT Class 12th Page No. 46]

168. (2)

The patients are given substances called biological response modifiers such as α-interferon which activates their immune system and helps in destroying the tumor.

[New NCERT Class 12th Page No. 142]

169. (3)



Given above diagram is of Copper -T IUDs.

[New NCERT Class 12th Page No. 44]

170. (1)

- Emergency contraceptive pills within 72 hours of coitus prevents conception.
- The Government of India legalised MTP in 1971.
- Nearly 45 to 50 million MTPs are performed in a year all over the world. .

[New NCERT Class 12th Page No. 46]

171. (2)

The cell-mediated immunity inside the human body is carried out by T-lymphocytes.

[New NCERT Class 12th Page No. 135]

172. (3)

- Hashish is obtained from resin of *Cannabis sativa.*
- Morphine leads to delusions and disturbed emotions.

[New NCERT Class 12th Page No. 142, 143]

173. (3)

IUDs increase phagocytosis of sperms within the uterus and the Cu ions released suppress sperm motility and the fertilising capacity of sperms.

[New NCERT Class 12th Page No. 44]

174. (2)

- The zygote or early embryos (with up to 8 blastomeres) could then be transferred into the fallopian tube (ZIFT–zygote intra fallopian transfer) and embryos with more than 8 blastomeres, into the uterus (IUT – intra uterine transfer), to complete its further development.
- Embryos formed by *in-vivo* fertilisation (fusion of gametes within the female) also could be used for such transfer to assist those females who cannot conceive.

[New NCERT Class 12th Page No. 48]

175. (3)

HIV	Retrovirus
Spermicidal cream	Increase contraceptive efficiency
ICSI	Sperms are directly injected into the ovum
Vaults	Entry of sperm through cervix is blocked

[New NCERT Class 12th Page No. 118]

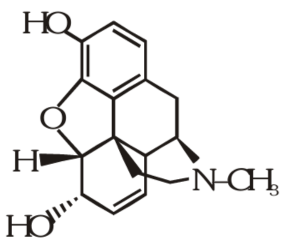
176. (4)

The discovery of blood circulation by William Harvey using experimental method and the demonstration of normal body temperature in persons with blackbile using thermometer disproved the 'good humor' hypothesis of health.

[New NCERT Class 12th Page No. 129]

177. (4)

Structure of morphine is



[New NCERT Class 12th Page No. 142]

178. (3)

Tobacco contains a large number of chemical substances including nicotine, an alkaloid. Nicotine stimulates adrenal gland to release adrenaline and nor-adrenaline into blood circulation, both of which raise blood pressure and increase heart rate.

[New NCERT Class 12th Page No. 144]

179. (2)

Heroin, commonly called smack, is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. This is obtained by acetylation of morphine.

[New NCERT Class 12th Page No. 179]

180. (4)

- Injection of dead or inactivated pathogens causes passive immunity.
- Passive immunity does not provide a permanent cure for disease.

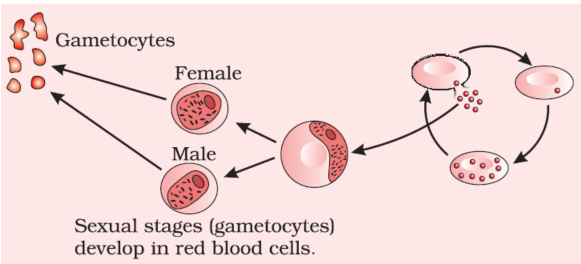
[New NCERT Class 12th Page No. 136]

181. (1)

After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix.

[New NCERT Class 12th Page No. 138]

182. (3)



[New NCERT Class 12th Page No. 132]

183. (2)

The zygote or early embryos (with up to 8 blastomeres) could then be transferred into the fallopian tube (ZIFT–zygote intra fallopian transfer).

[New NCERT Class 12th Page No. 42]

184. (2)

- Health does not simply mean 'absence of disease' or 'physical fitness'.
- The exaggerated response of the immune system to certain antigens present in the environment is called allergy. The substances to which such an immune response is produced are called allergens.

[New NCERT Class 12th Page No. 130]

185. (4)

Symptoms of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing.

[New NCERT Class 12th Page No. 137]

186. (2)

A contraceptive pill prevents ovulation by inhibiting release of FSH & LH.

[New NCERT Class 12th Page No. 45]

187. (4)

- Cancer causing viruses called oncogenic viruses modify the host's genome.
- Cancer detection is based on biopsy, bone marrow tests and histopathological studies of the tissue and blood.

[New NCERT Class 12th Page No. 141]

188. (4)

The progeny viruses released in the blood attack other helper T-lymphocytes. This is repeated leading to a progressive decrease in the number of helper T-lymphocytes in the body of the infected person.

[New NCERT Class 12th Page No. 140]

189. (1)

Nirodh' is a popular brand of condom for the male.

[New NCERT Class 12th Page No. 44]

190. (3)

- STIs are a major threat to a healthy society.
- Incidences of STIs are reported to be very high among persons in the age group of 15-24 years.

[New NCERT Class 12th Page No. 47]

191. (3)

Tetanus and cholera are caused by bacteria.

[New NCERT Class 12th Page No. 130]

192. (1)

- Gametocytes are produced in the red blood cells (RBCs) of humans, not in the liver, and they multiply through sexual reproduction in the mosquito.
- Human RBCs are ruptured and released a toxin haemozoin.

[New NCERT Class 12th Page No. 131]

193. (2)

The hormone releasing IUDs are Progestasert, LNG-20.

[New NCERT Class 12^{th} Page No. 44]

194. (3)

Allergy	IgE
Polio	Non- infectious
Widal test	Typhoid fever
ELISA test	AIDS

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195. (3)

Grafted kidney may be rejected in a patient due to cell mediated immune response.

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196. (3)

Wuchereria the filarial worms cause a slowly developing chronic inflammation of the organs in which they live for many years, usually the lymphatic vessels of the lower limbs and the disease is called elephantiasis or filariasis. The genital organs are also often affected.

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197. (1)

The primary lymphoid organs are bone marrow and thymus.

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198. (1)

Even in cases of snakebites, the injection which is given to the patients, contain preformed antibodies against the snake venom. This type of immunisation is called passive immunisation.

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199. (3)

GIFT – gamete intra fallopian transfer.

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200. (3)

IgE antibodies responsible for production of allergy.

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